**SOA Survey**

# Deep Learning Approaches for Decoding Cerebellar Purkinje Cell Dynamics

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**Abstract**

This survey studies how deep learning can be used to decode cerebellar Purkinje cell activity into meaningful representations of motor control, an important problem because these cells form the sole output of the cerebellum and are critical for understanding movement and motor learning. we reviewed several papers spanning three main sub-areas: reliable detection of rare spike events, spatio-temporal decoding of behavior from neural populations, and interpretability of coordinated neural activity. Prior work shows that deep models such as CNNs and LSTMs can improve spike detection and enable basic behavioral decoding, but most approaches remain limited to coarse tasks like binary classification and do not capture richer motor variables such as movement deceleration or behavioral intent. This reveals a gap between physiological evidence of coordinated, temporally precise neural dynamics and existing computational models that fail to fully exploit these structures. To address this, we propose to investigate attention-based spatio-temporal architectures that can learn which neurons and temporal patterns are most informative, with the goal of improving decoding of fine-grained motor behaviors from Purkinje cell activity.

## 1. Problem Definition

The task involves translating raw spike data from the Purkinje cells of a mammal into meaningful representations of motor control and coordination. In essence, this means taking discrete high dimensional data sourced from Electrophysiology recordings of live mammals and building deep learning models which can classify the data and therefore infer some relevant information.

This problem is important because Purkinje cells form the sole output of the cerebellar cortex, which is the part of the brain responsible for motor control and motor learning. If we can reliably decode their activity, we can better understand how the brain controls movement and adapts motor behavior.

There are a number of challenges that, collectively, make this problem difficult to tackle. These include:

- **Data Scarcity**: High-quality neural recordings are limited and expensive to obtain, leading to small sample sizes that can cause overfitting.
- **Signal Sparsity**: Neural activity is inherently sparse; important events like complex spikes occur very rarely within a recording. This creates a problem where the model must distinguish rare, functional signals from vast amounts of background noise.
- **Data Complexity**: The data is noisy, high-dimensional, and varies across recording sessions and subjects. One might argue that the problem of CS detection is rather trivial; a simple voltage-threshold-based detection could easily solve this problem. However, no matter how well isolated a PC may be, there may be fluctuations of the raw signal being recorded, which can occur at different time scales (whether fast or due to gradual drifts in the position of the neuron relative to the electrode). These fluctuations necessarily modify the start and end times of detected events using simple voltage thresholds. Therefore, even when clean signals can sometimes allow simple detection with thresholds for some applications, the relevance of CSs in the field of cerebellum research extends beyond "mere detection." Precise characterization of CSs and their overall durations, as well as the characterization of their morphology, may matter a great deal for function.

- **Representation Difficulty**: The model has to represent both where the spikes come from spatially (which cell they are coming from) and when the spikes are happening. Thus, any architecture must be spatiotemporal aware.

## 2. Literature Review

### 2.1. Challenge 1: Reliable detection of rare and complex Purkinje cell events

A central challenge is the reliable identification of Purkinje cell spike types, especially complex spikes, which are not only rare but also variable in form. The field cares about this problem because any downstream decoding or modeling pipeline depends on the quality of spike detection at the input stage. If complex spikes are missed or mislabeled, then later analyses of behavior decoding become unreliable.

Markanday et al. [2] address this challenge by introducing a deep neural network for automated complex spike detection, using both action potential and local field potential signals to improve event classification. Their method shows that deep learning can outperform traditional PCA-based and template-matching approaches, reducing the need for labor-intensive manual annotation and making large-scale pre-processing more feasible. It does not attempt to infer or decode motor behavior from these signals. It focuses on accurate event detection, but does not address how these detected spikes can be used to model or explain motor control.

### 2.2. Challenge 2: Decoding behavior from spatio-temporal Purkinje cell population activity

A second major challenge is learning how Purkinje cell activity patterns map onto behavior. Unlike the previous challenge, which focuses on accurately detecting spike events, this problem assumes that spikes have already been identified and instead asks how these signals can be used to infer motor function. This is more difficult than simple spike classification because the goal is not just to detect events, but to understand how patterns across multiple neurons and time steps relate to movement variables such as timing, coordination, or deceleration.

Bina et al. (2025) laid the groundwork for this decoding task by utilizing gradient boosting (XGBoost) to predict licking behavior from neural activity, demonstrating that Purkinje cell simple spike (SS) activity-and to a lesser extent complex spike (CS) signals-carry predictive information. However, their approach relied on handcrafted features, such as firing rates and inter-spike interval statistics, which may struggle to capture the non-linear temporal dependencies present in high-dimensional spike data.

Zeeshan et al. (2026) address this challenge by proposing deep spatio-temporal architectures for decoding Purkinje cell activity during rhythmic licking behavior. They evaluate both a convolutional neural network (PSN-CNN), which captures spatial relationships across electrodes, and a recurrent model (PSN-LSTM), which models temporal dependencies in spike sequences. While their models successfully capture spatial relationships across electrodes, they focus on binary classification (licking vs. non-licking) rather than modeling finer aspects of motor intent or deceleration.

To bridge this gap, Zhang et al. (2025) demonstrate that recurrent spatio-temporal models can decode higher-level behavioral intent. By utilizing a Bi-directional LSTM (BiLSTM) to capture context from both forward and backward temporal sequences, they successfully forecasted the directional choices (intent) of mice based on sparse spiking activity. This suggests that deep learning can extract ”intent” from population dynamics, even when the underlying neural signal is sparse.

However, the work by Zhang et al. remains limited to a simplified binary decision task (Left vs. Right) and does not evaluate whether these architectures can generalize across different subjects or sessions. Furthermore, because their model was not tested on cerebellar Purkinje cell data, it remains unknown if attention-based models can accurately decode continuous motor variables, such as movement deceleration, or distinguish between complex, multi-class behavioral intents like grooming and feeding.

### 2.3. Challenge 3: Interpreting coordinated Purkinje cell activity through attention-based models

A third challenge is moving beyond accurate prediction to understanding how coordinated Purkinje cell activity encodes meaningful aspects of motor behavior.

Zhang et al. (2025) address this challenge in part by introducing a channel-attention BiLSTM architecture, which assigns importance weights to individual neurons and improves decision decoding performance. Their results show that a subset of neurons contributes disproportionately to behavioral outcomes, and that attention mechanisms can help identify these task-relevant units. However, the decoding task remains relatively coarse, focusing on binary decision outcomes rather than richer behavioral variables.

Physiological studies provide further evidence that such coordination is critical. Hage et al. (2025) demonstrate that Purkinje cells encode movement deceleration, suggesting a role in predictive motor control, while Sedaghat-Nejad et al. (2022) show that synchronous spiking across populations carries behaviorally relevant information. These findings indicate that meaningful signals arise not only from individual neurons, but from temporally precise coordination across populations.

Although attention-based models can identify neurons that contribute to prediction, they have primarily been applied to coarse behavioral tasks and do not yet capture the coordinated spatio-temporal dynamics required to decode richer motor signals such as movement deceleration or behavioral intent (e.g., grooming versus feeding).

**Open Problems (flagged by the field:)**

- **Cross-session and cross-species generalisation**: A primary challenge remains in developing architectures that can generalize across different recording sessions and species, specifically moving from mice to marmoset cerebellar data. (Zhang et al., 2025; Zeeshan et al., 2026)
- **Decoding Multi-alternative and Complex Behaviors**: Most current models are restricted to simplified binary tasks. There is a significant gap in scaling these architectures to decode complex behaviors that involve multiple alternatives or intricate value calculations. (Zhang et al., 2025)

## 3. Research Gap and Proposed Direction

Existing work in this domain highlights several important but unresolved challenges. First, while deep learning methods such as those proposed by Zeeshan et al. (2026) demonstrate that Purkinje cell activity can be used to decode behavior, they are typically limited to coarse behavioral tasks (e.g., lick vs. no-lick) and do not capture finer aspects of motor control such as movement deceleration or behavioral intent. Second, physiological studies such as Hage et al. (2025) and Sedaghat-Nejad et al. (2022) show that Purkinje cells encode meaningful motor signals through temporally precise activity and population-level coordination, including signals related to stopping and movement control. However, current deep learning models do not explicitly incorporate or decode these richer behavioral variables, creating a gap between physiological insight and computational modeling.

Among these, the most critical and tractable gap is the lack of models that can decode fine-grained motor signals - such as deceleration and behavioral intent - while capturing spatio-temporal structure in neural population activity. Existing approaches tend to simplify the behavioral task.

We therefore propose to investigate attention-based spatio-temporal models that can learn which neurons and temporal patterns are most informative for these behaviors.

**Research Question (formal: 1-3 lines):** Can attention-based spatio-temporal neural architectures improve the decoding of Purkinje cell activity by capturing coordinated and temporally precise neural dynamics, enabling prediction of both movement deceleration and behavioral intent (e.g., grooming vs. feeding) from neural recordings?

## 4. Benchmark, Evaluation, and Data Landscape

**Datasets.**

Table 1: Key datasets in the field.

| Dataset | Size | Modality | Notes |
|---|---|---|---|
| Bina et al. Dataset | 84 Purkinje cells, 32 mice | Electrophysiology and Video | Lick detection and tongue tracking |
| Hage et al. Dataset | 284 Purkinje cells, 3 marmosets (840,787 licks) | Electrophysiology and Video | Targeted tongue movements |

**Evaluation Protocol.**

The community primarily uses evaluation protocols that are especially designed to address the fact that neural data is highly imbalanced. Zeeshan et al. rely on F1-scores and Balanced Accuracy (BAC) to ensure their models accurately identify licking behavior rather than simply achieving high accuracy by predicting the majority "no-lick" class. Similarly, to evaluate the decoding of behavioral intent, Zhang et al. utilize Accuracy and Area Under Curve (AUC) to validate their model's ability to predict "Left vs. Right" choices from sparse spiking activity.

**Compute Context.**

Key papers in this survey relied on standard CPU-based workstations for classical methods, with GPU support (GeForce GTX 1080Ti) used for DeepLabCut pose estimation in Hage et al. and for training deep models in Zeeshan et al.

## Experimental Design Considerations

It is essential to benchmark performance against a diverse set of baselines from classical machine learning to domain specific deep learning methods. First, we include classical baselines, particularly XGBoost and SVMs. Beating this baseline is important to evaluate the necessity of deep spatio-temporal architectures. Second, we incorporate PurkinjeSpikeNet architectures (PSN-CNN and PSN-LSTM) proposed in recent work by Zeeshan et al. (2026). To ensure a fair comparison, these architectures would be adapted from binary to multi-label prediction by modifying the output layer to produce independent probabilities per label. Outperforming these models is critical, as they represent the current state-of-the-art in modeling spatio-temporal dependencies in Purkinje cell activity.

A meaningful ablation study would involve isolating the specific components of the data and the model architecture to determine their relative contributions to decoding accuracy. The following variables will be isolated:

- **Temporal Window Size**: Compare shorter windows (100–150ms) against the 200ms windows used by Zeeshan et al. (2026). Their window size was calibrated for binary lick detection; finer behavioral distinctions such as grooming versus feeding may require a different temporal resolution.
- **Simple Spikes vs. Complex Spikes**: Train the model on simple spikes alone, then with both spike types combined. This is directly motivated by Hage et al. (2025), who show that SS and CS carry complementary information

One of the most significant challenges in this domain is class imbalance, as the majority of recording windows typically correspond to "no-lick" events. Without careful management, a model might achieve deceptively high accuracy by simply predicting the majority class, and fail to capture the actual underlying neural dynamics of motor behavior. To mitigate this, it is essential to follow the precedent set by Zeeshan et al. (2026), who prioritized Balanced Accuracy and F1-scores over raw accuracy to ensure the model maintains high sensitivity to the rarer, behaviorally relevant spike patterns.

## 5. Annotated Reading List

- **Bina et al. (2025)**
  Established the foundational dataset and behavioral paradigm for targeted tongue movements in mice. They utilized gradient boosting (XGBoost) and SHAP analysis to show that Purkinje cell simple spike rates and complex spike timing can predict lick onsets. This work serves as the

primary data source and baseline for binary lick detection.

- **Markanday et al. (2020)**
  Proposes a convolutional neural network for automated detection of complex spikes in Purkinje cell recordings. While effective for preprocessing, the work is limited to spike detection and does not address behavioral decoding.

- **Sedaghat-Nejad et al. (2022)**
  Investigates synchronous spiking across Purkinje cell populations during movement control. The study shows that coordinated activity across neurons plays a key role in encoding motor behavior. However, the work is primarily physiological and does not provide computational models for decoding these patterns.

- **Hage et al. (2025)**
  Demonstrates that Purkinje cells encode movement deceleration during tongue movements, suggesting a role in predictive motor control. The study provides strong biological evidence linking neural activity to motor function. However, it does not propose a deep learning framework for decoding these signals.

- **Zhang et al. (2025)**
  Proposes a channel-attention BiLSTM model (CA-BiLSTM) for decoding decision-making behavior from neural spike data across brain regions. The attention mechanism identifies neurons most relevant to behavioral outcomes, improving both accuracy and interpretability. However, the decoding task remains limited to simple decision outputs and does not address richer behavioral variables such as motor control or intent.

- **Zeeshan et al. (2026)**
  Introduces deep spatio-temporal models (CNN and LSTM) for decoding Purkinje cell activity during licking behavior. The approach represents spike data as structured matrices and shows improved performance over classical methods. However, the task is limited to coarse behavioral decoding.

## References


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[2] A. Markanday, J. Bellet, M. E. Bellet, J. Inoue, Z. M. Hafed, and P. Thier. Using deep neural networks to detect complex spikes of cerebellar Purkinje cells. *Journal of Neurophysiology*, 123(6):2217–2234, 2020.

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## 6. Preliminary Implementation Resources

- https://github.com/CVLABLUMS/psn
  This is a reference implementation of the PSN-CNN and PSN-LSTM architectures from the Zeeshan et al. (2026) paper.

- https://github.com/openai/CLIP
  A multimodal model pretrained on large scale image text pairs used by Zeeshan et al. (2026) for comparison with deep learning baselines.

- https://github.com/ZhangYH321/CA-BiLSTM/tree/master

All data and code used for running experiments, model fitting, and plotting in the Zhang et al.(2025) paper.

- https://github.com/jobellet/detect_CS
  The implementation of the CNN-based complex spike detection algorithm from Markanday et al. (2020).

- https://github.com/DeepLabCut/DeepLabCut
  An open-source toolbox for pose estimation used by Hage et al. (2025) and Bina et al. (2025) to extract the high-fidelity kinematic variables required for training behavioral decoding models